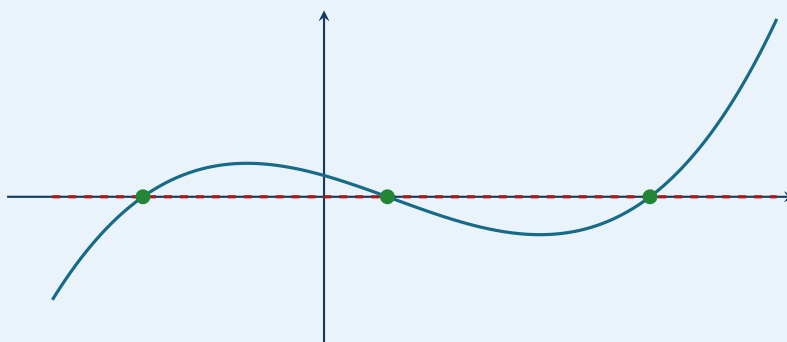


Applications of Derivatives

Increasing and Decreasing Functions, Extrema, Concavity, Inflection Points, and Curve Analysis

Complete Colored English LaTeX Edition

A comprehensive reconstruction of the full 26-page source file, including definitions, procedures, worked examples, special cases, graph interpretation, parameter problems, practice questions, and answer keys.



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AP Calculus AB / BC

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Source Coverage Map

Pages	Content incorporated in this English edition
1–2	Definitions of increasing, decreasing, and constant functions; absolute and local extrema; critical numbers; the relationship between the sign of f' and the behavior of f ; the first derivative test; a complete solution workflow.
3–9	Polynomial, radical, rational, trigonometric, absolute-value, piecewise, discontinuous, and step-function examples; endpoint effects; repeated critical numbers; points where f' does not exist; local versus absolute extrema.
10	Parameter problems involving prescribed critical points, local extrema, and reconstruction of polynomial functions from conditions.
11–14	Concavity, tangent-line interpretation, points of inflection, horizontal inflection, angle of inflection, second-derivative sign charts, and special cases where $f'' = 0$ or is undefined but no inflection occurs.
15–16	Parameter problems involving inflection points and tangent lines; the second derivative test; inconclusive cases; comparison with the first derivative test.
17–20	Reading information from graphs of f , f' , and f'' ; reconstructing and sketching functions; comparing f , f' , and f'' at labeled points.
21–26	Comprehensive assessment material: multiple-choice questions, structured free-response problems, graph-based analysis, parameter problems, monotonicity, extrema, concavity, inflection, and answer keys.

How to use this book

Blue boxes introduce definitions, green boxes state theorems, gold boxes give procedures, red boxes warn about common errors, purple boxes highlight deeper ideas, and gray boxes contain practice. Worked examples are fully explained and the final part contains a comprehensive assessment and teacher key.

1 Learning Objectives

By the end of this lesson, students will be able to:

- determine intervals on which a function is increasing, decreasing, or constant;
- identify critical numbers and explain why endpoints and discontinuities require separate treatment;
- find local and absolute extrema using the first derivative test and the closed-interval method;
- analyze polynomial, rational, radical, trigonometric, absolute-value, and piecewise functions;
- determine concavity and points of inflection using f'' ;
- calculate the angle of inflection from the slope of the tangent line;
- apply the second derivative test and recognize when it is inconclusive;
- infer the behavior of f from graphs of f' , and infer concavity from graphs of f'' ;
- reconstruct or sketch a function from derivative information;
- solve parameter problems involving prescribed extrema or inflection points.

Part I: Increasing, Decreasing, and Extrema

2 Increasing, Decreasing, and Constant Functions

Increasing on an interval

A function f is **increasing** on an interval I if, whenever $x_1 < x_2$ in I ,

$$f(x_1) < f(x_2).$$

Graphically, the curve rises as we move from left to right.

Decreasing on an interval

A function f is **decreasing** on an interval I if, whenever $x_1 < x_2$ in I ,

$$f(x_1) > f(x_2).$$

Graphically, the curve falls as we move from left to right.

Constant on an interval

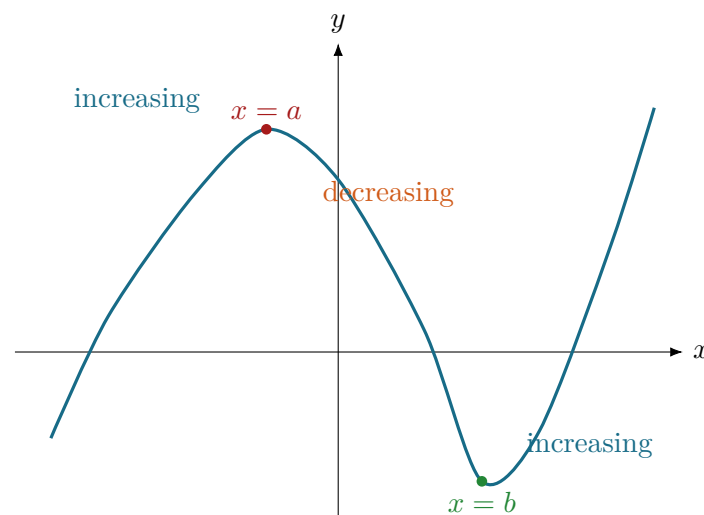
A function f is **constant** on an interval I if

$$f(x_1) = f(x_2)$$

for every pair $x_1, x_2 \in I$.

Important

Intervals of increase and decrease are normally reported as **open intervals**. Endpoints may be mentioned separately when the domain is closed, but a derivative sign chart is divided at critical numbers, discontinuities, and domain endpoints.



3 Absolute and Local Extrema

Absolute maximum and minimum

Let f be defined on a domain D .

- $f(c)$ is an **absolute maximum** if $f(c) \geq f(x)$ for every $x \in D$.
- $f(c)$ is an **absolute minimum** if $f(c) \leq f(x)$ for every $x \in D$.

The words *global maximum* and *global minimum* mean the same thing.

Local maximum and minimum

- $f(c)$ is a **local maximum** if $f(c) \geq f(x)$ for all x sufficiently close to c .
- $f(c)$ is a **local minimum** if $f(c) \leq f(x)$ for all x sufficiently close to c .

A local extremum only compares nearby points; an absolute extremum compares every point in the entire domain.

Extreme Value Theorem

If f is continuous on a **closed interval** $[a, b]$, then f attains both an absolute maximum and an absolute minimum on $[a, b]$.

Important

The Extreme Value Theorem guarantees extrema only on a closed, bounded interval and only when the function is continuous. It does not say where the extrema occur.

3.1 Endpoint conventions

A closed endpoint may be an absolute extremum. Depending on the convention adopted in a course, it may also be called a one-sided local extremum. In AP Calculus work, always state clearly whether you are using one-sided neighborhoods at endpoints.

3.2 Constant functions

If $f(x) = k$ on an interval, then every point in the interval has both the maximum value and the minimum value k . Under a non-strict local definition, every point is simultaneously a local maximum and local minimum.

4 Critical Numbers

Critical number

A number c in the domain of f is a **critical number** if

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

The point $(c, f(c))$ is a critical point on the graph.

Important

A number at which f is not defined is **not** a critical number. It may still divide the domain into separate intervals and must appear on the sign chart as a discontinuity.

Fermat's Theorem

If f has a local extremum at an interior point c and $f'(c)$ exists, then $f'(c) = 0$. The converse is false: $f'(c) = 0$ does not automatically imply an extremum.

Smooth turning point	$f'(c) = 0$ and the derivative changes sign.
Horizontal non-turning point	$f'(c) = 0$ but the derivative does not change sign; no extremum.
Corner or cusp	$f'(c)$ does not exist, but $f(c)$ exists; the point can be an extremum.
Vertical tangent	$f'(c)$ may be infinite or undefined; the point may or may not be an extremum.
Hole / vertical asymptote	$f(c)$ does not exist, so c is not a critical number.
Endpoint	Not usually classified as a critical number, but must be tested for absolute extrema on a closed interval.

5 The Derivative and Monotonicity

Monotonicity Test

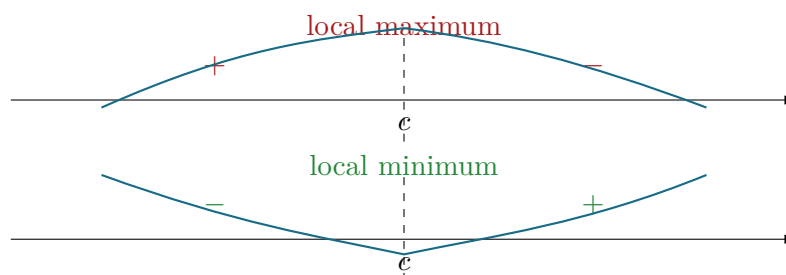
Let f be differentiable on an interval I .

- If $f'(x) > 0$ for every $x \in I$, then f is increasing on I .
- If $f'(x) < 0$ for every $x \in I$, then f is decreasing on I .
- If $f'(x) = 0$ for every $x \in I$, then f is constant on I .

First Derivative Test

Suppose c is a critical number of f .

- If f' changes from positive to negative at c , then f has a local maximum at c .
- If f' changes from negative to positive at c , then f has a local minimum at c .
- If f' has the same sign on both sides of c , then f has no local extremum at c .

**Complete workflow for increasing/decreasing intervals and extrema**

1. Find the domain of f .
2. Compute $f'(x)$.

3. Find all critical numbers: solve $f'(x) = 0$ and include points where f' is undefined but f is defined.
4. Add domain discontinuities and endpoints to the number line.
5. Test the sign of f' on each interval.
6. State intervals of increase and decrease.
7. Use sign changes to classify local extrema.
8. For absolute extrema on $[a, b]$, evaluate f at every interior critical number and at both endpoints.

6 Worked Examples: First Derivative Analysis

Example 1: A quadratic

Analyze $f(x) = x^2 - 4x + 2$.

Solution

$$f'(x) = 2x - 4 = 2(x - 2).$$

The only critical number is $x = 2$. Since $f'(x) < 0$ for $x < 2$ and $f'(x) > 0$ for $x > 2$,

$$f \text{ decreases on } (-\infty, 2), \quad f \text{ increases on } (2, \infty).$$

Thus f has a local and absolute minimum at $x = 2$:

$$f(2) = 4 - 8 + 2 = -2.$$

There is no maximum on $\mathbb{R}$.

Example 2: Closed interval and endpoints

Find the absolute extrema of $f(x) = 4 - x^2$ on $[-1, 2]$.

Solution

$$f'(x) = -2x = 0 \implies x = 0.$$

Evaluate the critical number and endpoints:

$$f(-1) = 3, \quad f(0) = 4, \quad f(2) = 0.$$

Therefore the absolute maximum is 4 at $x = 0$, and the absolute minimum is 0 at $x = 2$. The function increases on $(-1, 0)$ and decreases on $(0, 2)$.

Example 3: A cubic with two critical numbers

Analyze $f(x) = x^3 - 3x^2 + 1$.

Solution

$$f'(x) = 3x^2 - 6x = 3x(x - 2).$$

Critical numbers: 0 and 2. The sign pattern is $+, -, +$. Hence f increases on $(-\infty, 0)$ and $(2, \infty)$, and decreases on $(0, 2)$.

$$f(0) = 1 \quad (\text{local maximum}), \quad f(2) = 8 - 12 + 1 = -3 \quad (\text{local minimum}).$$

Example 4: Three critical numbers

Analyze

$$f(x) = \frac{1}{4}x^4 - 2x^2 + 1.$$

Solution

$$f'(x) = x^3 - 4x = x(x-2)(x+2).$$

Critical numbers: $-2, 0, 2$. The sign pattern is $-, +, -, +$. Therefore:decreasing on $(-\infty, -2) \cup (0, 2)$,increasing on $(-2, 0) \cup (2, \infty)$.Local minima occur at $x = \pm 2$ with value -3 . A local maximum occurs at $x = 0$ with value 1 .**Example 5: Repeated factors and a stationary non-extremum**Analyze $f(x) = (x+1)(x-2)^6$.**Solution**

$$f'(x) = (x-2)^5(7x+4).$$

Critical numbers are $x = -\frac{4}{7}$ and $x = 2$. The sign changes $+\rightarrow -$ at $-\frac{4}{7}$, so there is a local maximum there. The sign changes $- \rightarrow +$ at 2 , so there is a local minimum at $x = 2$, where $f(2) = 0$. The high multiplicity at $x = 2$ creates a very flat turning point.**Example 6: A critical number with no extremum**Analyze $f(x) = (x-1)^5$.**Solution**

$$f'(x) = 5(x-1)^4 \geq 0.$$

Although $x = 1$ is critical, the derivative is positive on both sides. Hence f is increasing on all of $\mathbb{R}$ and has no local extremum at $x = 1$.**Example 7: A semicircle**Analyze $f(x) = \sqrt{16-x^2}$ on $[-4, 4]$.**Solution**The domain is $[-4, 4]$ and

$$f'(x) = \frac{-x}{\sqrt{16-x^2}}.$$

The derivative is undefined at the endpoints and zero at $x = 0$. It is positive on $(-4, 0)$ and negative on $(0, 4)$. Thus f increases on $(-4, 0)$ and decreases on $(0, 4)$. The absolute maximum is 4 at $x = 0$; the absolute minimum is 0 at $x = \pm 4$.**Example 8: Two disconnected branches**Analyze $f(x) = \sqrt{x^2-16}$ on $(-\infty, -4] \cup [4, \infty)$.

Solution

$$f'(x) = \frac{x}{\sqrt{x^2 - 16}}.$$

The function decreases on $(-\infty, -4)$ and increases on $(4, \infty)$. Both endpoints $x = -4$ and $x = 4$ give the absolute minimum value 0. There is no absolute maximum.

Example 9: A cusp and undefined derivative

Analyze

$$f(x) = (x^2 - 4)^{2/3}.$$

Solution

$$f'(x) = \frac{4x}{3\sqrt[3]{x^2 - 4}}.$$

Candidates: $x = 0$ from $f' = 0$, and $x = \pm 2$ where f' is undefined but f exists. The derivative sign pattern is

$$- \mid_{-2} + \mid_0 - \mid_2 +.$$

Therefore $x = \pm 2$ are local and absolute minima with value 0, while $x = 0$ is a local maximum with value $\sqrt[3]{16}$.

Example 10: Rational function

Analyze $f(x) = \frac{x}{1 + x^2}$.

Solution

$$f'(x) = \frac{1 - x^2}{(1 + x^2)^2}.$$

Critical numbers: $x = \pm 1$. Since the denominator is always positive, the sign is controlled by $1 - x^2$. The function decreases on $(-\infty, -1)$ and $(1, \infty)$ and increases on $(-1, 1)$.

$$f(-1) = -\frac{1}{2} \quad \text{local/absolute minimum}, \quad f(1) = \frac{1}{2} \quad \text{local/absolute maximum}.$$

Example 11: Trigonometric function on a closed interval

Analyze $f(x) = 2 \cos x$ on $[0, 2\pi]$.

Solution

$$f'(x) = -2 \sin x.$$

Interior critical number: $x = \pi$. The function decreases on $(0, \pi)$ and increases on $(\pi, 2\pi)$.

$$f(\pi) = -2 \quad \text{absolute minimum}, \quad f(0) = f(2\pi) = 2 \quad \text{absolute maxima}.$$

Example 12: A bounded trigonometric function

Analyze $f(x) = \sin(2x)$ on $[0, \frac{\pi}{2}]$.

Solution

$$f'(x) = 2 \cos(2x) = 0 \implies x = \frac{\pi}{4}.$$

The derivative changes from positive to negative, so the absolute maximum is 1 at $x = \pi/4$. The endpoint values are both 0, so the absolute minimum is 0 at $x = 0$ and $x = \pi/2$.

Example 13: Derivative zero without a change of monotonicity

Analyze $f(x) = x + \sin x$ on $\left[0, \frac{3\pi}{2}\right]$.

Solution

$$f'(x) = 1 + \cos x \geq 0,$$

with equality only at $x = \pi$. The sign does not change, so $x = \pi$ is not an extremum. The function is increasing over the whole interval. The absolute minimum occurs at $x = 0$, and the absolute maximum at $x = 3\pi/2$.

Example 14: Multiple trigonometric extrema

Analyze $f(x) = \sin^4 x$ on $[0, 2\pi]$.

Solution

$$f'(x) = 4 \sin^3 x \cos x.$$

Critical numbers are $0, \pi/2, \pi, 3\pi/2, 2\pi$ (including endpoints for the closed-interval check). Absolute maxima of 1 occur at $\pi/2$ and $3\pi/2$. Absolute minima of 0 occur at $0, \pi, 2\pi$.

Example 15: Absolute value

Analyze $f(x) = x|x - 3|$.

Solution

Write

$$f(x) = \begin{cases} 3x - x^2, & x < 3, \\ x^2 - 3x, & x \geq 3. \end{cases}$$

Then

$$f'(x) = \begin{cases} 3 - 2x, & x < 3, \\ 2x - 3, & x > 3. \end{cases}$$

Candidates are $x = 3/2$ and the corner $x = 3$. The derivative changes $+$ $\rightarrow$ $-$ at $3/2$, giving a local maximum $f(3/2) = 9/4$. At $x = 3$ the derivative changes $-$ $\rightarrow$ $+$, giving a local minimum $f(3) = 0$.

7 Special Cases and Common Traps

7.1 Discontinuities

A discontinuity can separate intervals of monotonicity, but it is not a critical number unless the function is actually defined at that x -value. A local extremum can occur at a jump point when the function value is greater or less than nearby values, even if the derivative does not change sign in the usual continuous sense.

Important

Do not decide local extrema only from arrows on a sign chart when the function is discontinuous. Compare the actual function value at the discontinuity with nearby values.

7.2 Piecewise functions

For a piecewise function:

1. differentiate each formula on its own interval;
2. analyze the joining points separately;
3. check continuity and compare one-sided values;
4. include endpoints in the absolute-extrema check.

Piecewise example

Let

$$f(x) = \begin{cases} x^2 + 1, & x < 1, \\ 4 - x, & 1 \leq x \leq 4, \\ (x - 4)^2, & x > 4. \end{cases}$$

Find the intervals of increase/decrease and all local extrema.

Solution

For $x < 1$, $f'(x) = 2x$, so the function decreases on $(-\infty, 0)$ and increases on $(0, 1)$. At $x = 0$, f has a local minimum $f(0) = 1$. For $1 < x < 4$, $f'(x) = -1$, so it decreases. For $x > 4$, $f'(x) = 2(x - 4) > 0$, so it increases. At $x = 1$, $f(1) = 3$, while the left-hand values approach 2; therefore $x = 1$ is a one-sided local maximum. At $x = 4$, $f(4) = 0$ and nearby values are larger, so $x = 4$ is a local minimum.

7.3 Radicals and domain endpoints

For an even root $f(x) = \sqrt{g(x)}$, first solve $g(x) \geq 0$. Points where $g(x) = 0$ often make f' undefined and may be endpoint extrema. For an odd root such as $\sqrt[3]{g(x)}$, there is no sign restriction on $g(x)$, but the derivative may still be undefined when $g(x) = 0$.

7.4 Repeated roots of f'

If a factor of f' has even multiplicity, the sign of f' usually does not change across that root; if it has odd multiplicity, the sign usually changes. Always verify with a sign chart.

8 Closed-Interval Method

Finding absolute extrema on $[a, b]$

1. Verify that f is continuous on $[a, b]$.
2. Find all critical numbers in (a, b) .
3. Evaluate f at each critical number and at a and b .
4. The largest value is the absolute maximum; the smallest is the absolute minimum.

Closed-interval comparison

Find the absolute extrema of $f(x) = x^3 - 3x$ on $[-2, 3]$.

Solution

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1),$$

so the critical numbers are $x = -1$ and $x = 1$.

x	-2	-1	1	3
$f(x)$	-2	2	-2	18

The absolute maximum is 18 at $x = 3$. The absolute minimum is -2 , attained at both $x = -2$ and $x = 1$.

9 Parameter Problems Involving Extrema

Key Insight

If (a, b) is a smooth interior extremum of f , then two conditions are available:

$$f(a) = b, \quad f'(a) = 0.$$

These conditions produce equations for unknown parameters. They are necessary, but a sign test or second derivative test may still be needed to verify the type of extremum.

Parameter Example 1

If $f(x) = ax^2 - 12x$ has a critical point at $x = 2$, find a .

Solution

$$f'(x) = 2ax - 12,$$

so $f'(2) = 4a - 12 = 0$, giving $\boxed{a = 3}$.

Parameter Example 2

If $f(x) = ax^2 - bx + 5$ has a local maximum at $(1, 4)$, find a and b .

Solution

The point lies on the curve:

$$a - b + 5 = 4 \implies a - b = -1.$$

The tangent is horizontal:

$$f'(x) = 2ax - b, \quad f'(1) = 2a - b = 0.$$

Solving gives $a = 1$ and $b = 2$. But $f''(x) = 2 > 0$, so the point is actually a local minimum, not a maximum. Therefore the stated maximum condition is inconsistent. This illustrates why verification matters.

Parameter Example 3: Reconstructing a cubic

Find a cubic $f(x) = ax^3 + bx^2 + cx + d$ having a critical point at $(0, 5)$ and a local minimum at $(1, 3)$.

Solution

From $(0, 5)$: $d = 5$ and $c = 0$. From $(1, 3)$:

$$a + b + 5 = 3 \implies a + b = -2.$$

Since $x = 1$ is critical,

$$f'(x) = 3ax^2 + 2bx + c, \quad 3a + 2b = 0.$$

Solving gives $a = 4$, $b = -6$, so

$$f(x) = 4x^3 - 6x^2 + 5.$$

Since $f''(1) = 24 - 12 = 12 > 0$, the point is indeed a local minimum.

Part II: Concavity and Points of Inflection

10 Concavity

Concave up

A differentiable function is **concave up** on an interval if its graph lies above its tangent lines there. Equivalently, the slopes $f'(x)$ are increasing.

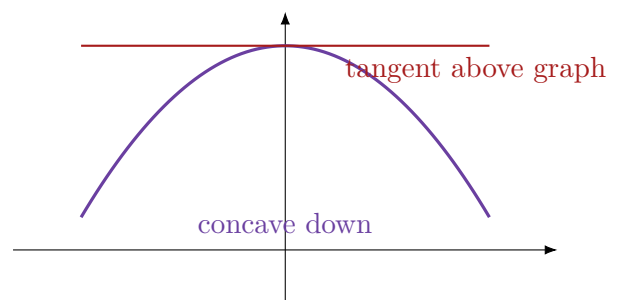
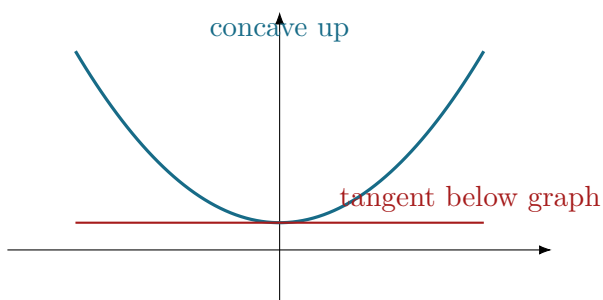
Concave down

A differentiable function is **concave down** on an interval if its graph lies below its tangent lines there. Equivalently, the slopes $f'(x)$ are decreasing.

Second Derivative and Concavity

Suppose f'' exists on an interval I .

- If $f''(x) > 0$ on I , then f is concave up on I .
- If $f''(x) < 0$ on I , then f is concave down on I .



11 Points of Inflection

Point of inflection

A point $(c, f(c))$ is a **point of inflection** if:

1. f is continuous at c , and
2. the concavity of f changes at c .

Important

The equation $f''(c) = 0$ is only a way to find a candidate. It does not prove that an inflection point exists. The sign of f'' must change. Also, a discontinuity cannot be an inflection point because the point is not on a continuous curve.

Complete workflow for concavity and inflection points

1. Find the domain of f .
2. Compute $f''(x)$.
3. Find candidates by solving $f''(x) = 0$ and including points where f'' is undefined but f is defined.
4. Add discontinuities and endpoints to the number line.
5. Test the sign of f'' on each interval.
6. State intervals of concavity.
7. A candidate is an inflection point only if f is continuous there and f'' changes sign.

11.1 Angle of inflection

If $(c, f(c))$ is an inflection point and the tangent exists, the angle of inflection θ is measured counterclockwise from the positive x -axis and satisfies

$$\tan \theta = f'(c), \quad 0 \leq \theta < \pi.$$

If $f'(c) = 0$, the point is a **horizontal inflection point** and $\theta = 0$. If $f'(c) < 0$, choose the angle in the second quadrant:

$$\theta = \pi + \arctan(f'(c)).$$

12 Worked Examples: Concavity and Inflection**Example 16: A cubic**

Analyze the concavity of $f(x) = x^3 - 3x^2 + 1$.

Solution

$$f''(x) = 6x - 6 = 6(x - 1).$$

Thus $f'' < 0$ for $x < 1$ and $f'' > 0$ for $x > 1$. The function is concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$. Since the sign changes,

$$(1, f(1)) = (1, -1)$$

is a point of inflection.

Example 17: Two inflection points

Analyze $f(x) = x^4 - 4x^3 + 1$ on $[-2, 4]$.

Solution

$$f''(x) = 12x^2 - 24x = 12x(x - 2).$$

The sign pattern is $+, -, +$ across 0 and 2. Therefore f is concave up on $(-2, 0)$ and $(2, 4)$, and concave down on $(0, 2)$. The inflection points are

$$(0, 1), \quad (2, -15).$$

Example 18: $f'' = 0$ but no inflection

Analyze the concavity of $f(x) = x^4 + 5$.

Solution

$$f''(x) = 12x^2 \geq 0.$$

Although $f''(0) = 0$, the second derivative is positive on both sides of 0. The graph is concave up on all of $\mathbb{R}$, and there is no inflection point.

Example 19: Disconnected domain

Analyze $f(x) = x + \frac{1}{x}$.

Solution

$$f''(x) = \frac{2}{x^3}.$$

The function is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$. Even though the sign changes at 0, there is no inflection point because f is not defined at $x = 0$.

Example 20: Undefined second derivative without sign change

Analyze $f(x) = 3 + (x + 4)^{2/3}$.

Solution

For $x \neq -4$,

$$f''(x) = -\frac{2}{9(x+4)^{4/3}} < 0.$$

The function is concave down on both $(-\infty, -4)$ and $(-4, \infty)$. The second derivative is undefined at -4 , but the concavity does not change, so there is no inflection point.

Example 21: Upper semicircle

Analyze $f(x) = \sqrt{16 - x^2}$.

Solution

For $-4 < x < 4$,

$$f''(x) = -\frac{16}{(16 - x^2)^{3/2}} < 0.$$

Hence the graph is concave down throughout $(-4, 4)$ and has no inflection point.

Example 22: Trigonometric concavity

Analyze

$$f(x) = 1 + \cos x - \cos^2 x, \quad 0 \leq x \leq 2\pi.$$

Solution

$$f'(x) = \sin x(2 \cos x - 1),$$

$$f''(x) = 2 \cos(2x) - \cos x.$$

Set $2 \cos(2x) - \cos x = 0$. Using $\cos(2x) = 2 \cos^2 x - 1$ gives

$$4 \cos^2 x - \cos x - 2 = 0,$$

so

$$\cos x = \frac{1 \pm \sqrt{33}}{8}.$$

These values generate the inflection candidates in $[0, 2\pi]$. A sign chart for f'' determines the alternating concavity intervals. This is a typical case where an exact symbolic equation is preferable to decimal guessing.

Example 23: Absolute value and concavity

Analyze $f(x) = |x^2 - 4|$.

Solution

$$f(x) = \begin{cases} 4 - x^2, & |x| < 2, \\ x^2 - 4, & |x| \geq 2. \end{cases}$$

Thus $f'' = -2$ on $(-2, 2)$ and $f'' = 2$ on $(-\infty, -2) \cup (2, \infty)$. The graph changes concavity at $x = \pm 2$, and it is continuous there. Therefore $(\pm 2, 0)$ are inflection points, even though f' is undefined at the corners.

13 Parameter Problems Involving Inflection**Inflection Parameter Example 1**

Let $f(x) = ax^3 + 6x^2 + bx$ have an inflection point at $(1, 0)$. Find a and b .

Solution

Because the point lies on the curve,

$$a + 6 + b = 0. \quad (1)$$

Also,

$$f''(x) = 6ax + 12,$$

so $f''(1) = 0$ gives $6a + 12 = 0$, hence $a = -2$. From (1), $b = -4$. Thus

$$f(x) = -2x^3 + 6x^2 - 4x.$$

Since $f''(x) = 12(1 - x)$ changes sign at $x = 1$, the point is genuinely an inflection point.

Inflection Parameter Example 2

A cubic $f(x) = ax^3 + bx^2 + cx + d$ passes through $(0, 5)$ and has a horizontal inflection point at $(1, 3)$. Find the family of all such cubics.

Solution

The conditions are

$$f(0) = 5 \Rightarrow d = 5,$$

$$f(1) = 3 \Rightarrow a + b + c = -2,$$

$$f'(1) = 0 \Rightarrow 3a + 2b + c = 0,$$

$$f''(1) = 0 \Rightarrow 6a + 2b = 0 \Rightarrow b = -3a.$$

Substituting gives $c = 3a$ and then $a - 3a + 3a = -2$, so $a = -2$. Hence

$$f(x) = -2x^3 + 6x^2 - 6x + 5.$$

Inflection angle

Suppose $(2, -5)$ is an inflection point and $f'(2) = -1/\sqrt{3}$. Find the angle of inflection.

Solution

The reference angle is $\pi/6$. Since the slope is negative and $0 \leq \theta < \pi$, the tangent angle lies in Quadrant II:

$$\theta = \frac{5\pi}{6}.$$

14 The Second Derivative Test

Second Derivative Test

Suppose $f'(c) = 0$.

- If $f''(c) > 0$, then f has a local minimum at c .
- If $f''(c) < 0$, then f has a local maximum at c .
- If $f''(c) = 0$ or does not exist, the test is inconclusive; use the first derivative test or another argument.

Second derivative test on a cubic

Use the second derivative test for $f(x) = x^3 - 3x^2 + 1$.

Solution

$f'(x) = 3x(x - 2)$, so the critical numbers are 0 and 2. Since $f''(x) = 6x - 6$,

$$f''(0) = -6 < 0 \Rightarrow \text{local maximum at } (0, 1),$$

$$f''(2) = 6 > 0 \Rightarrow \text{local minimum at } (2, -3).$$

Second derivative test on a rational function

Use the second derivative test for $f(x) = x + 4/x$.

Solution

$$f'(x) = 1 - \frac{4}{x^2} = 0 \implies x = \pm 2, \quad f''(x) = \frac{8}{x^3}.$$

Thus $f''(-2) = -1 < 0$, so $(-2, -4)$ is a local maximum. Also $f''(2) = 1 > 0$, so $(2, 4)$ is a local minimum.

Inconclusive case

Use the second derivative test for $f(x) = x^4$.

Solution

$f'(0) = 0$ and $f''(0) = 0$, so the second derivative test is inconclusive. However,

$$f'(x) = 4x^3$$

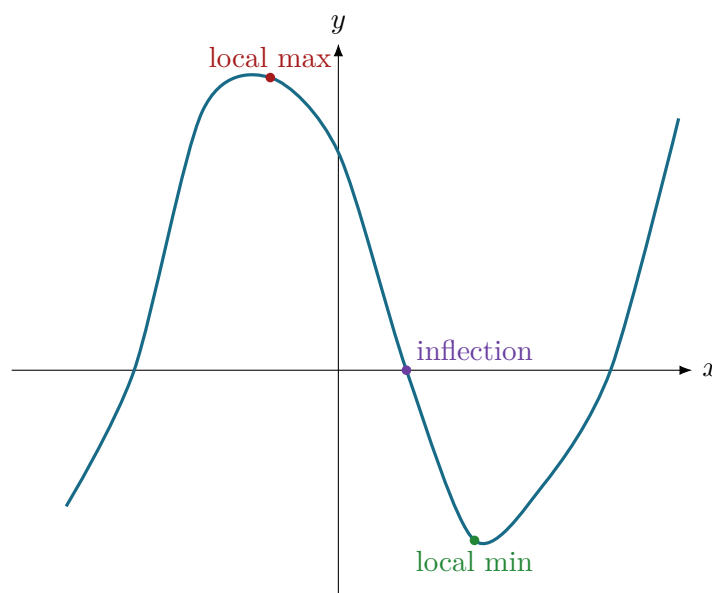
changes from negative to positive at 0, so f has a local and absolute minimum at $(0, 0)$.

Part III: Reading and Constructing Graphs

15 When the Given Graph Is f

From the graph of f itself:

- rising portions indicate increase;
- falling portions indicate decrease;
- peaks and valleys indicate local extrema;
- bending upward/downward indicates concavity;
- a change of concavity at a continuous point indicates an inflection point;
- the slope of the tangent gives $f'(x)$.



16 When the Given Graph Is f'

Reading f from f'

- $f' > 0$ means f is increasing.
- $f' < 0$ means f is decreasing.
- A sign change $+\rightarrow -$ in f' gives a local maximum of f .
- A sign change $-\rightarrow +$ in f' gives a local minimum of f .
- Where f' is increasing, $f'' > 0$ and f is concave up.
- Where f' is decreasing, $f'' < 0$ and f is concave down.
- A local extremum of f' often corresponds to an inflection point of f , provided f is continuous and the monotonicity of f' changes.

Graph of f' : a parabola

Suppose the graph of f' is the parabola $f'(x) = (x - 2)^2 - 4$, with zeros at $x = 0$ and $x = 4$.

Solution

$f' > 0$ on $(-\infty, 0) \cup (4, \infty)$ and $f' < 0$ on $(0, 4)$. Therefore f increases, then decreases, then increases. It has a local maximum at $x = 0$ and a local minimum at $x = 4$. The graph of f' decreases on $(-\infty, 2)$ and increases on $(2, \infty)$, so f is concave down then concave up, with an inflection point at $x = 2$.

Graph of f' : comparisons

At labeled points P, Q, R, S, T on a graph of f , compare the signs of f' and f'' using the slope and concavity of the curve.

Solution

At a point where the curve is rising and concave down, $f' > 0$ and $f'' < 0$. At a local maximum, $f' = 0$ and usually $f'' < 0$. At a falling point with concave up shape, $f' < 0$ and $f'' > 0$. At an inflection point, f'' changes sign and may equal 0 if it exists.

17 When the Given Graph Is f''

Reading f from f''

- $f'' > 0$ means f is concave up.
- $f'' < 0$ means f is concave down.
- A sign change in f'' gives an inflection point of f if f is continuous.
- A graph of f'' alone usually does not determine where f is increasing or decreasing, because the sign of f' is unknown.

Graph of f'' : a line

Suppose $f''(x) = 2 - x$ on $[0, 4]$.

Solution

$f'' > 0$ for $x < 2$ and $f'' < 0$ for $x > 2$. Hence f is concave up on $(0, 2)$ and concave down on $(2, 4)$, with an inflection point at $x = 2$. If we are additionally told that $f'(2) = 0$, then f has neither a local maximum nor minimum there because the concavity changes from up to down while the tangent is horizontal: it is a horizontal inflection point.

18 Sketching from Derivative Information**Sketching a possible graph of f**

1. Mark critical numbers and discontinuities.
2. Use the sign of f' to draw rising/falling portions.
3. Mark local extrema where the sign of f' changes.
4. Use the sign of f'' to bend the graph correctly.
5. Mark inflection points where concavity changes.
6. Incorporate any known points, endpoint values, asymptotes, or tangent slopes.
7. Remember that many different vertical shifts can satisfy the same derivative information.

Sketching challenge

Sketch a function f satisfying all of the following:

- f is increasing on all of $\mathbb{R}$;
- f is concave down for $x < 2$ and concave up for $x > 2$;
- $(2, 1)$ is a horizontal inflection point.

One possible answer is a vertical shift and scaling of $(x - 2)^3$, such as $f(x) = (x - 2)^3 + 1$.

Part IV: Comprehensive Practice**19 Master Practice Set A: Monotonicity and Extrema**

For each function, determine the domain, critical numbers, intervals of increase/decrease, and all local and absolute extrema that exist.

A1. $f(x) = x^2 - 4x + 2$.

A2. $f(x) = 4 - x^2$ on $[-1, 2]$.

A3. $f(x) = x^3 - 3x^2 + 1$.

A4. $f(x) = x^3 - 3x$ on $[0, 3]$.

A5. $f(x) = x^3 - 3x$ on $(0, 3)$.

A6. $f(x) = x^3 - 3x$ on $[0, 3)$.

A7. $f(x) = x^3 + x + 1$.

A8. $f(x) = \frac{1}{4}x^4 - 2x^2 + 1$.

A9. $f(x) = x^5 - \frac{5}{3}x^3$.

A10. $f(x) = (x - 1)^5$.

A11. $f(x) = (x + 1)(x - 2)^6$.

A12. $f(x) = \sqrt{2x - 4}$.

A13. $f(x) = \sqrt{16 - x^2}$.

A14. $f(x) = \sqrt{x^2 - 16}$.

A15. $f(x) = (x^2 - 4)^{2/3}$.

A16. $f(x) = \frac{x}{1 + x^2}$.

A17. $f(x) = x^{1/3}(x-2)^{2/3}$.

A18. $f(x) = x + \frac{4}{x}$.

A19. $f(x) = \frac{x^2}{1+x^2}$.

A20. $f(x) = \frac{x}{1+x^2}$ on $[-2, 2]$.

A21. $f(x) = 2 \cos x$ on $[0, 2\pi]$.

A22. $f(x) = \sin(2x)$ on $[0, \pi/2]$.

A23. $f(x) = \sqrt{3}x + \cos x$.

A24. $f(x) = x + \sin x$ on $[0, 3\pi/2]$.

A25. $f(x) = \sin^4 x$ on $[0, 2\pi]$.

A26. $f(x) = \sqrt{\cos x}$ on $[-\pi/2, \pi/2]$.

A27. $f(x) = \sin x + \cos x$ on $[0, 2\pi]$.

A28. $f(x) = \begin{cases} x^2 - 1, & x \leq 1, \\ 3 - x, & x > 1. \end{cases}$

A29. $f(x) = \begin{cases} 2x, & x < 1, \\ 3, & 1 \leq x < 3, \\ x - 1, & x \geq 3. \end{cases}$

A30. $f(x) = \begin{cases} x^2 + 5, & -2 \leq x < 3, \\ 11 - 2x, & 3 \leq x \leq 7. \end{cases}$

A31. $f(x) = x|x-3|$.

20 Master Practice Set B: Concavity and Inflection

For each function, determine intervals of concavity and all points of inflection. Where requested, find the angle of inflection.

B1. $f(x) = x^3 - \frac{3}{2}x^2$.

B2. $f(x) = x^4 - 4x^3 + 1$ on $[-2, 4]$.

B3. $f(x) = 6x^5 - x^6$.

B4. $f(x) = x^4 - 3x^3 + 5$.

B5. $f(x) = 2x - 5\sqrt{x} + 1$.

B6. $f(x) = x + \frac{1}{x}$.

B7. $f(x) = 3 + (x+4)^{2/3}$.

B8. $f(x) = \sqrt{16 - x^2}$.

B9. $f(x) = x^{2/3}$.

B10. $f(x) = 1 + \cos x - \cos^2 x$ on $[0, 2\pi]$.

B11. $f(x) = \frac{1}{2}x^2 + 2 \cos x$ on $[0, 2\pi]$.

B12. $f(x) = |x^2 - 4|$.

B13. $f(x) = \begin{cases} x^3, & x < 0, \\ x^2, & x \geq 0. \end{cases}$

B14. $f(x) = \begin{cases} x^2, & x < 2, \\ 4x - 4, & x \geq 2. \end{cases}$

21 Graph-Based Practice

G1: Graph of f'

The derivative f' is positive on $(-4, -1)$, negative on $(-1, 2)$, and positive on $(2, 5)$. It is decreasing on $(-4, 0)$ and increasing on $(0, 5)$.

- State the intervals where f increases and decreases.
- Classify the critical numbers -1 and 2 .
- State the concavity of f .
- Identify any inflection point of f .

G2: Graph of f''

Suppose $f''(x)$ is positive on $(-\infty, -3)$, negative on $(-3, 1)$, and positive on $(1, \infty)$.

- State the concavity intervals of f .

- (b) Identify the possible inflection points.
- (c) Explain why no conclusion about increasing/decreasing intervals can be made without additional information.

G3: Comparison of derivatives

At a point P on a smooth graph, the curve is decreasing but becoming less steep as x increases. Determine the signs of $f'(P)$ and $f''(P)$.

G4: Horizontal inflection

A function satisfies $f'(2) = 0$, $f''(x) < 0$ for $x < 2$, and $f''(x) > 0$ for $x > 2$. What type of point is $(2, f(2))$? Explain why the second derivative test for extrema is inconclusive.

22 Multiple-Choice Assessment

Choose the best answer.

- If $f'(x) = |x - 3| - 2$, the critical numbers of f are
(A) 1 only (B) 5 only (C) 1, 5 (D) 3.
- If $f'(x) = \sqrt{9 - x^2}$ and the domain of f is $[-3, 3]$, then f is
(A) decreasing on $(-3, 3)$ (B) increasing on $(-3, 3)$ (C) constant on $(-3, 3)$ (D) neither increasing nor decreasing.
- The function $f(x) = x^3 - 3x$ has a local maximum at
(A) $x = -1$ (B) $x = 0$ (C) $x = 1$ (D) $x = 3$.
- If $f'(x) = (x - 2)^2(x + 1)$, then f has
(A) a local max at -1 only (B) a local min at -1 only (C) local extrema at -1 and 2 (D) no local extrema.
- If $f'(x) > 0$ and $f''(x) < 0$ on an interval, then f is
(A) increasing and concave up (B) increasing and concave down (C) decreasing and concave up (D) decreasing and concave down.
- A point c with $f''(c) = 0$ is necessarily
(A) a local maximum (B) a local minimum (C) an inflection point (D) none of these necessarily.
- If $f'(2) = 0$ and $f''(2) = 7$, then f has at $x = 2$
(A) a local maximum (B) a local minimum (C) an inflection point (D) no conclusion.
- If $f'(2) = 0$ and $f''(2) = 0$, then

- (A) f has a maximum (B) f has a minimum sive.
(C) f has an inflection (D) the test is inconclu-
9. The function $f(x) = x^{2/3}$ has at $x = 0$
- (A) a smooth local maximum (B) a cusp local (C) an inflection point (D) no extremum.
minimum
10. For $f(x) = x + 1/x$, $x = 0$ is
- (A) a critical number (B) an inflection point a local extremum.
(C) a discontinuity but not a critical number (D)
11. If f' changes from negative to positive at c , then $f(c)$ is
- (A) a local maximum (B) a local minimum (C) an inflection point (D) an endpoint.
12. If f'' changes from positive to negative at c and f is continuous at c , then $(c, f(c))$ is
- (A) a local minimum (B) a local maximum (C) an inflection point (D) not on the graph.
13. The absolute maximum of $f(x) = 4 - x^2$ on $[-1, 2]$ is
- (A) 0 (B) 3 (C) 4 (D) 5.
14. The absolute minimum of $f(x) = \sqrt{16 - x^2}$ on $[-4, 4]$ is
- (A) -4 (B) 0 (C) 4 (D) does not exist.
15. If $f'(x) > 0$ for every x except $f'(1) = 0$, then
- (A) f must have a maximum at 1 (B) f must (C) f is increasing and may have no extremum at
have a minimum at 1 1 (D) f is decreasing.
16. If $f''(x) > 0$ on all of $\mathbb{R}$ and $f'(3) = 0$, then $x = 3$ is
- (A) an absolute minimum (B) an absolute max- (C) an inflection point (D) a discontinuity.
imum
17. For $f(x) = \sin x + \cos x$ on $[0, 2\pi]$, the maximum value is
- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\pi/4$.
18. The point $x = \pi$ for $f(x) = x + \sin x$ is
- (A) a local maximum (B) a local minimum critical.
(C) a critical number with no extremum (D) not
19. If $f''(x) = 6x$, then f is concave down on
- (A) $(-\infty, 0)$ (B) $(0, \infty)$ (C) all $\mathbb{R}$ (D) nowhere.
20. If $f'(c) = m < 0$, the tangent angle θ in $[0, \pi)$ is

- (A) $\arctan m$ (B) $\pi + \arctan m$ (C) $-\arctan m$ (D) $\pi - \arctan m$.

21. If the graph of f' has a local minimum at $x = 4$, then f may have at $x = 4$

- (A) an inflection point (B) necessarily a local minimum (C) necessarily a local maximum (D) necessarily a discontinuity.

22. If f'' is positive on both sides of c and $f''(c) = 0$, then c

- (A) must be an inflection point (B) cannot be a critical number (C) is not an inflection point based on concavity (D) must be a maximum.

23. The function $f(x) = |x^2 - 4|$ is concave down on

- (A) $(-2, 2)$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) all $\mathbb{R}$.

24. If f is continuous on $[a, b]$, then the Extreme Value Theorem guarantees

- (A) exactly one maximum (B) exactly one minimum (C) at least one absolute maximum and minimum (D) no critical numbers.

23 Free-Response Assessment

FR1: Complete polynomial analysis

For $f(x) = x^4 - 4x^3 + 2$:

- (a) find all critical numbers;
- (b) determine intervals of increase/decrease;
- (c) find all local extrema;
- (d) determine concavity intervals and inflection points;
- (e) sketch a possible graph showing all key features.

FR2: Closed interval

Find the absolute maximum and minimum of $f(x) = x^3 - 6x^2 + 9x + 1$ on $[0, 5]$.

FR3: Radical function

Analyze $f(x) = \sqrt{x^2 - 9}$ on its natural domain. Include the domain, monotonicity, extrema, concavity, and any inflection points.

FR4: Rational function

For $f(x) = \frac{x^2}{1 + x^2}$, determine the intervals of increase/decrease, extrema, concavity, and inflection points.

FR5: Parameter and extremum

Find a, b, c so that $f(x) = x^3 + ax^2 + bx + c$ has a local maximum at $(0, 4)$ and a local minimum at $(2, 0)$.

FR6: Parameter and inflection

Find a and b if $f(x) = ax^3 + bx^2 - 6x + 1$ has a horizontal inflection point at $x = 1$.

FR7: Piecewise behavior

Let

$$f(x) = \begin{cases} x^2 + 2x, & x < 0, \\ 2 - x, & 0 \leq x < 2, \\ (x - 2)^2, & x \geq 2. \end{cases}$$

Determine all intervals of increase/decrease and all local extrema, paying special attention to $x = 0$ and $x = 2$.

FR8: Trigonometric analysis

Analyze $f(x) = \sin x + \cos x$ on $[0, 2\pi]$: find all local and absolute extrema, intervals of monotonicity, concavity intervals, and inflection points.

FR9: Derivative graph data

Suppose f' is negative on $(-\infty, -2)$, positive on $(-2, 1)$, negative on $(1, 4)$, and positive on $(4, \infty)$. Also, f' is increasing on $(-\infty, 0)$, decreasing on $(0, 3)$, and increasing on $(3, \infty)$. Determine all local extrema and inflection points of f , and state all intervals of increase, decrease, concavity up, and concavity down.

FR10: Second derivative graph data

Suppose $f''(x) = (x + 2)(x - 1)(x - 4)$. Determine the concavity of f and all possible inflection points. If $f'(1) = 0$, explain what can and cannot be concluded about an extremum at $x = 1$.

FR11: Construct a function

Construct a polynomial of least possible degree that is increasing on all of $\mathbb{R}$, has a horizontal inflection point at $(2, 5)$, and is concave down for $x < 2$ and concave up for $x > 2$.

FR12: Proof and reasoning

Suppose $f'(x) = g'(x)$ for every x in an interval I . Prove that $f(x) - g(x)$ is constant on I . Then prove that if f and g are increasing differentiable functions on I , their sum $f + g$ is increasing on I .

Part V: Answer Key and Selected Solutions

24 Multiple-Choice Key

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	C	B	A	B	B	D	B	D	B	C	B	C

Question	13	14	15	16	17	18	19	20	21	22	23	24
Answer	C	B	C	A	B	C	A	B	A	C	A	C

25 Selected Free-Response Solutions

FR1

$$f(x) = x^4 - 4x^3 + 2.$$

$$f'(x) = 4x^2(x - 3).$$

Critical numbers: 0, 3. Since the factor x^2 does not change sign, $f' < 0$ for $x < 3$ (except $f'(0) = 0$) and $f' > 0$ for $x > 3$. Thus f decreases on $(-\infty, 3)$ and increases on $(3, \infty)$; $x = 0$ is not an extremum and $x = 3$ is a local/absolute minimum with $f(3) = -25$.

$$f''(x) = 12x(x - 2).$$

Concave up on $(-\infty, 0) \cup (2, \infty)$ and concave down on $(0, 2)$. Inflection points: $(0, 2)$ and $(2, -14)$.

FR2

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3). \text{ Evaluate at } 0, 1, 3, 5:$$

$$f(0) = 1, \quad f(1) = 5, \quad f(3) = 1, \quad f(5) = 21.$$

Absolute maximum: 21 at $x = 5$. Absolute minimum: 1 at $x = 0$ and $x = 3$.

FR4

$$\text{For } f(x) = x^2/(1 + x^2),$$

$$f'(x) = \frac{2x}{(1 + x^2)^2}.$$

The function decreases on $(-\infty, 0)$ and increases on $(0, \infty)$, with an absolute minimum 0 at $x = 0$. Also

$$f''(x) = \frac{2(1 - 3x^2)}{(1 + x^2)^3}.$$

It is concave up on $(-1/\sqrt{3}, 1/\sqrt{3})$ and concave down outside this interval. Inflection points occur at

$$\left(\pm \frac{1}{\sqrt{3}}, \frac{1}{4}\right).$$

FR5

Let $f(x) = x^3 + ax^2 + bx + c$. At $(0, 4)$: $c = 4$ and $f'(0) = b = 0$. At $(2, 0)$:

$$8 + 4a + 4 = 0 \Rightarrow a = -3.$$

Then $f'(x) = 3x^2 - 6x = 3x(x - 2)$, so 0 is a local maximum and 2 a local minimum. Hence

$$\boxed{a = -3, \quad b = 0, \quad c = 4}.$$

FR6

$f'(x) = 3ax^2 + 2bx - 6$ and $f''(x) = 6ax + 2b$. Horizontal inflection at $x = 1$ gives

$$3a + 2b - 6 = 0, \quad 6a + 2b = 0.$$

Subtracting gives $3a + 6 = 0$, so $a = -2$ and $b = 6$. Since $f''(x) = 12(1 - x)$ changes sign at 1, the point is genuinely an inflection point.

FR8

$f'(x) = \cos x - \sin x = \sqrt{2} \cos(x + \pi/4)$. Critical numbers in $[0, 2\pi]$ are $\pi/4$ and $5\pi/4$. The maximum value is $\sqrt{2}$ at $\pi/4$ and the minimum value is $-\sqrt{2}$ at $5\pi/4$.

$$f''(x) = -\sin x - \cos x = -\sqrt{2} \sin(x + \pi/4).$$

Inflection points occur where $\sin x + \cos x = 0$, namely $x = 3\pi/4, 7\pi/4$, with corresponding function value 0.

FR11

The least-degree choice is cubic. A horizontal inflection at $x = 2$ is produced by $(x - 2)^3$, and the point $(2, 5)$ is obtained by adding 5:

$$f(x) = (x - 2)^3 + 5.$$

Then $f'(x) = 3(x - 2)^2 \geq 0$ and $f''(x) = 6(x - 2)$, giving the required monotonicity and concavity.

FR12

Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ on I . By the Mean Value Theorem, h is constant on I . If f and g are increasing differentiable functions, then $f'(x) \geq 0$ and $g'(x) \geq 0$ wherever the derivative criterion applies, so

$$(f + g)'(x) = f'(x) + g'(x) \geq 0.$$

Thus $f + g$ is increasing. The same conclusion also follows directly from the definition of increasing functions.

26 Concise Key for Master Set A

-
- A1** Decreasing $(-\infty, 2)$; increasing $(2, \infty)$; absolute min $(2, -2)$.
 - A2** Increasing $(-1, 0)$; decreasing $(0, 2)$; absolute max 4 at 0, absolute min 0 at 2.
 - A3** Increasing $(-\infty, 0) \cup (2, \infty)$; decreasing $(0, 2)$; local max $(0, 1)$; local min $(2, -3)$.
 - A4** $f' = 3(x - 1)(x + 1)$; on $[0, 3]$: decreasing $(0, 1)$, increasing $(1, 3)$; local/absolute min -2 at 1; absolute max 18 at 3.
 - A7** Increasing on $\mathbb{R}$; no extrema.
 - A8** Decreasing $(-\infty, -2) \cup (0, 2)$; increasing $(-2, 0) \cup (2, \infty)$; minima at ± 2 , local max at 0.
 - A9** $f' = 5x^2(x^2 - 1)$; increasing $(-\infty, -1) \cup (1, \infty)$; decreasing $(-1, 1)$; local max at -1 , local min at 1; $x = 0$ stationary but not an extremum.
 - A10** Increasing on $\mathbb{R}$; no extrema.
 - A12** Domain $[2, \infty)$; increasing; absolute min 0 at $x = 2$.
 - A13** Increasing $(-4, 0)$; decreasing $(0, 4)$; max 4 at 0, mins 0 at ± 4 .
 - A14** Decreasing $(-\infty, -4)$; increasing $(4, \infty)$; absolute mins at ± 4 .
 - A15** Local mins at ± 2 and local max at 0.
 - A16** Decreasing $(-\infty, -1) \cup (1, \infty)$; increasing $(-1, 1)$; min $-1/2$, max $1/2$.
 - A18** Critical ± 2 ; local max $(-2, -4)$ and local min $(2, 4)$.
 - A19** Decreasing $(-\infty, 0)$; increasing $(0, \infty)$; absolute min 0 at 0; horizontal asymptote $y = 1$.
 - A21** Absolute maxima 2 at $0, 2\pi$; absolute min -2 at π .

- A22** Absolute max 1 at $\pi/4$; absolute mins 0 at endpoints.
- A23** $f'(x) = \sqrt{3} - \sin x > 0$; increasing on $\mathbb{R}$; no extrema.
- A24** Increasing on the full interval; $x = \pi$ is stationary but not an extremum.
- A25** Maxima 1 at $\pi/2, 3\pi/2$; minima 0 at $0, \pi, 2\pi$.
- A26** Domain $[-\pi/2, \pi/2]$; increasing $(-\pi/2, 0)$, decreasing $(0, \pi/2)$; max 1 at 0, mins 0 at endpoints.
- A27** Max $\sqrt{2}$ at $\pi/4$; min $-\sqrt{2}$ at $5\pi/4$.
- A31** Local max $9/4$ at $3/2$ and local min 0 at 3.
-

27 Final Checklist

Before submitting any derivative-analysis problem, verify that you have:

- stated the domain;
- separated critical numbers from discontinuities and endpoints;
- shown a correct sign chart for f' or f'' ;
- used open intervals for monotonicity and concavity;
- evaluated function values when naming extrema or inflection points;
- tested endpoints for absolute extrema on closed intervals;
- checked continuity before declaring an inflection point;
- verified the type of a claimed extremum rather than relying only on $f'(c) = 0$;
- used the correct quadrant for a negative tangent slope when finding an angle of inflection.

End of Complete English Edition